

Unit 1: Design Process and Aerospace Missions

Quiz 1A: Design Process, Systems, and Mission Objectives

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- What is the engineering design process used for?
 - Making things look nice only
 - Solving problems within criteria and constraints
 - Avoiding testing
 - Choosing the first idea
- Which statement best describes a mission objective?
 - The final product name
 - What the design must accomplish
 - A list of all materials
 - A grading rubric
- Which mission objective is strongest?
 - Build a drone
 - Make a cool aircraft
 - Deliver a fragile sensor package safely to a remote location
 - Use cardboard and tape
- What is a subsystem?
 - A separate class assignment
 - A smaller functional part of a larger system
 - A random decoration
 - A failed prototype
- Which is an aerospace subsystem of a drone?
 - Lunch schedule
 - Propulsion system
 - Class period
 - Notebook title
- Why should engineers define the mission before designing?
 - To limit creativity
 - To make sure solutions target the actual need
 - To skip research
 - To guarantee one answer
- Which step usually comes before brainstorming?
 - Problem definition
 - Final presentation
 - Mass production
 - Decorating

8. What is iteration?
- | | |
|-----------------------------|--|
| A. Never changing a design | B. Improving a design after learning from evidence |
| C. Copying a design exactly | D. Submitting late work |

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|---|--|
| _____ Mission objective | A. A clear statement of what the solution must accomplish. |
| _____ System | B. A group of interacting parts working together. |
| _____ Subsystem | C. A smaller functional part of a larger system. |
| _____ Iteration | D. Improving a solution based on evidence or feedback. |
| _____ Engineering design process | E. A structured process for solving engineering problems. |

Part C: Short Response

1. Explain why “build a drone” is not a strong mission objective.

2. Describe one aerospace system and two subsystems within it.

Unit 1: Design Process and Aerospace Missions

Quiz 1B: Stakeholders, Criteria, Constraints, and Problem Statements

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- Who is a stakeholder?
A. Only the engineer
B. Anyone affected by the design
C. Only the teacher
D. Only the person paying
- Which is a criterion?
A. Must cost under ten dollars
B. Should protect the payload
C. Only use supplied materials
D. Must be done today
- Which is a constraint?
A. Should be reliable
B. Should be accurate
C. Limited material choices
D. Should be easy to recover
- What should a strong problem statement include?
A. Only the final solution
B. Who has the problem, what is needed, why it matters, and key limits
C. Only a picture
D. A list of team roles
- Why should a problem statement avoid choosing the final solution too early?
A. It prevents multiple possible solutions from being considered
B. It makes grading harder
C. It removes criteria
D. It avoids teamwork
- Which stakeholder might care about payload recovery?
A. Emergency responders
B. No one
C. Only the logo designer
D. Only the computer
- Criteria define...
A. What makes the design successful
B. What color the paper is
C. Who gets extra credit
D. Where the file is saved

8. Constraints define...
- | | |
|----------------------------------|---------------------------|
| A. Limits the design must follow | B. Only the best ideas |
| C. The final answer | D. The presentation theme |

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|--------------------------------|--|
| _____ Stakeholder | A. A person or group affected by a design. |
| _____ Criterion | B. A goal or feature used to judge success. |
| _____ Constraint | C. A limit or requirement the design must follow. |
| _____ Problem statement | D. A clear description of the need to be solved. |
| _____ Payload | E. The item or equipment the system carries for the mission. |

Part C: Short Response

1. Write one criterion and one constraint for a fragile sensor delivery mission.

2. Explain why local emergency responders are stakeholders in the mission.

Unit 1: Design Process and Aerospace Missions

Quiz 1C: Research, Brainstorming, Trade-Offs, Decision Matrices, and Design Review

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. Research in engineering should help students...
 - A. Copy the first design they find
 - B. Learn from existing solutions and identify useful features
 - C. Avoid original thinking
 - D. Skip documentation
2. What is a trade-off?
 - A. A perfect solution
 - B. A situation where improving one factor may make another worse
 - C. A type of sketch paper
 - D. A final grade
3. Why brainstorm many ideas?
 - A. To delay work
 - B. To explore possibilities before selecting a final concept
 - C. To avoid criteria
 - D. To make packets longer
4. What does a decision matrix compare?
 - A. Students against each other
 - B. Design concepts using criteria
 - C. Only colors
 - D. Only file names
5. The highest decision matrix score should be...
 - A. Considered along with evidence and team reasoning
 - B. Ignored completely
 - C. Automatically copied
 - D. Used without explanation
6. Which is useful design review content?
 - A. Mission objective, sketches, decision matrix, risks, and next steps
 - B. Only team names
 - C. Only a final picture
 - D. Only a joke slide

7. Why should engineers identify limitations?
 - A. To make the design look bad
 - B. To plan testing and improvement
 - C. To avoid grading
 - D. To remove criteria
8. What makes feedback useful?
 - A. Specific, kind, connected to criteria, and focused on improvement
 - B. Only positive
 - C. Only negative
 - D. Unrelated to evidence

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|------------------------------|--|
| _____ Research | A. Studying existing solutions to learn useful features and limitations. |
| _____ Brainstorming | B. Generating many possible ideas before judging them. |
| _____ Trade-off | C. A compromise between competing design goals. |
| _____ Decision matrix | D. A structured tool for comparing concepts against criteria. |
| _____ Design review | E. A presentation or meeting used to evaluate progress and next steps. |

Part C: Short Response

1. Explain why a decision matrix is better than simply voting for a favorite idea.

2. Describe one trade-off that might affect an aerospace payload delivery design.
